

TIC 380943251

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.720935 d
Epoch (t0) 1386.4490 BTJD
Duration 0.0580 d
Depth 7.667 % (76,670 ppm)
SNR 101.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 27.8 (gate ≥ 7.0)
Odd/even z-score 1.89 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 3 light-curve dip in this sector: 25.9 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 15.26 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 380941073
4.65 px away, dTmag +0.59 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

